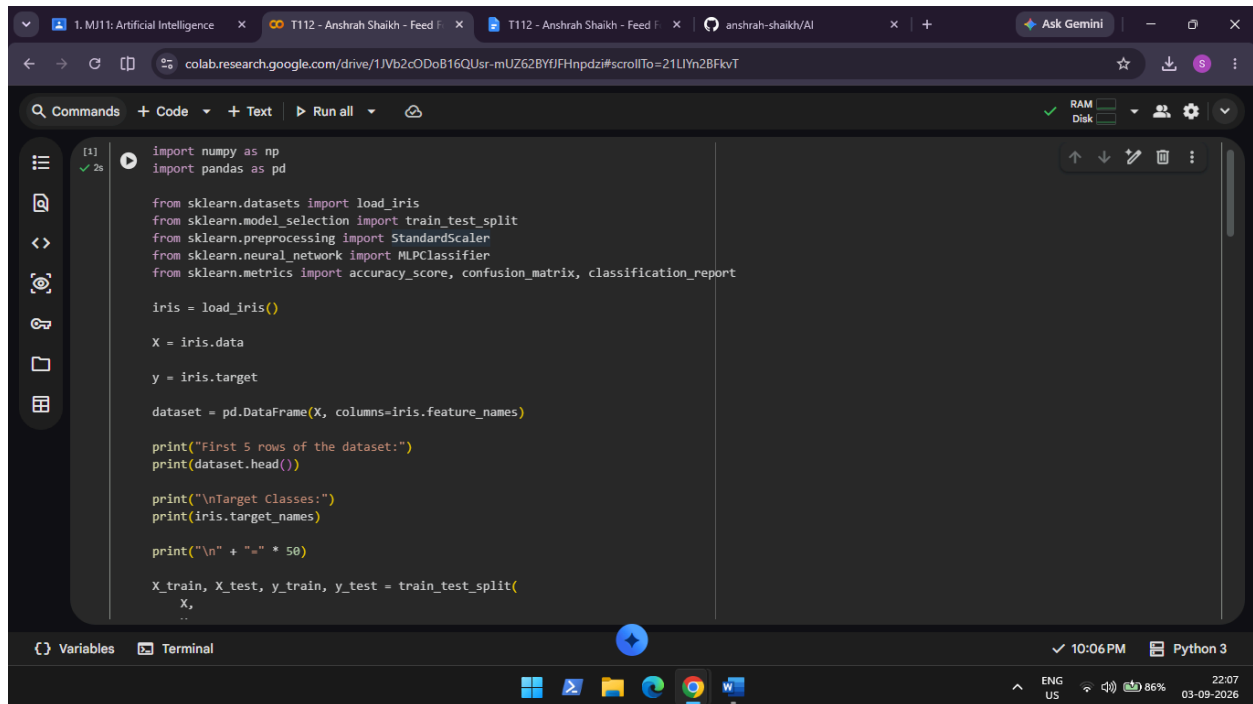


MVLU COLLEGE

Aim: Feed Forward Backpropagation Neural Network

- Implement the Feed Forward Backpropagation algorithm to train a neural network.
- Use a given dataset to train the neural network for a specific task.
- Evaluate the performance of the trained network on test data.

Input:



```
[1] ✓ 2s
import numpy as np
import pandas as pd

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

iris = load_iris()

X = iris.data
y = iris.target

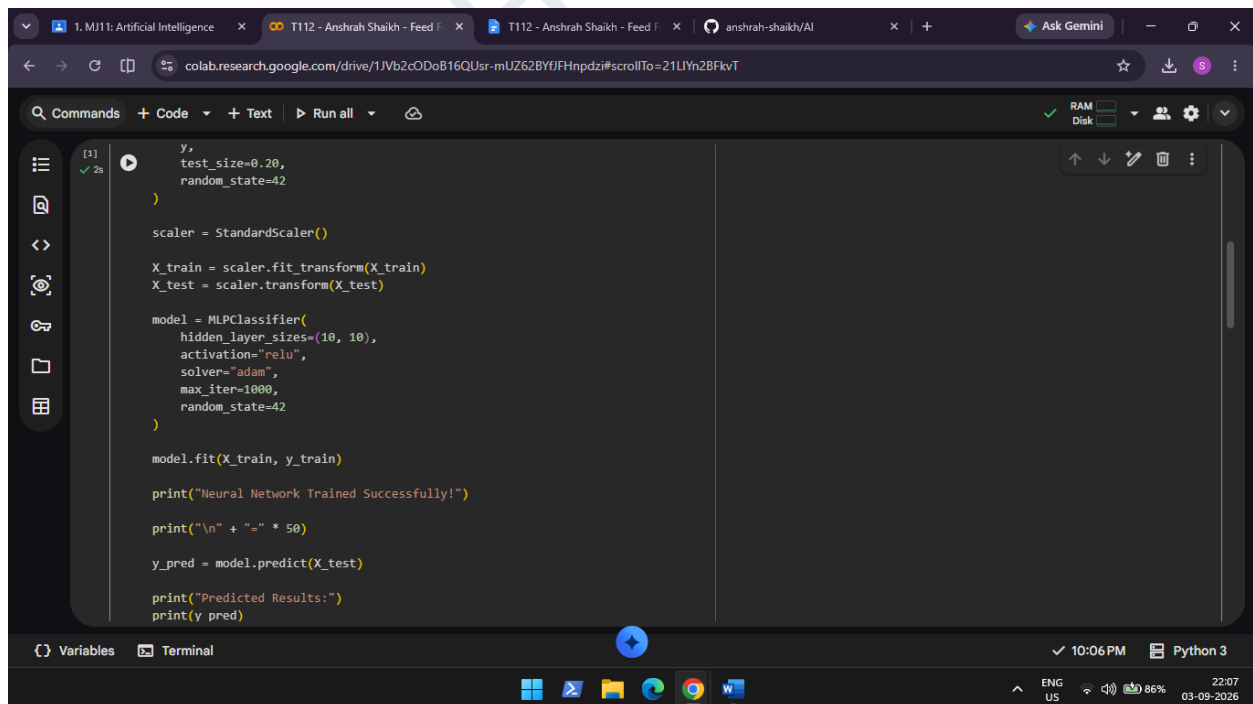
dataset = pd.DataFrame(X, columns=iris.feature_names)

print("First 5 rows of the dataset:")
print(dataset.head())

print("\nTarget Classes:")
print(iris.target_names)

print("\n" + "=" * 50)

X_train, X_test, y_train, y_test = train_test_split(
    X,
```



```
    y,
    test_size=0.20,
    random_state=42
)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10, 10),
    activation="relu",
    solver="adam",
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)

print("Neural Network Trained Successfully!")

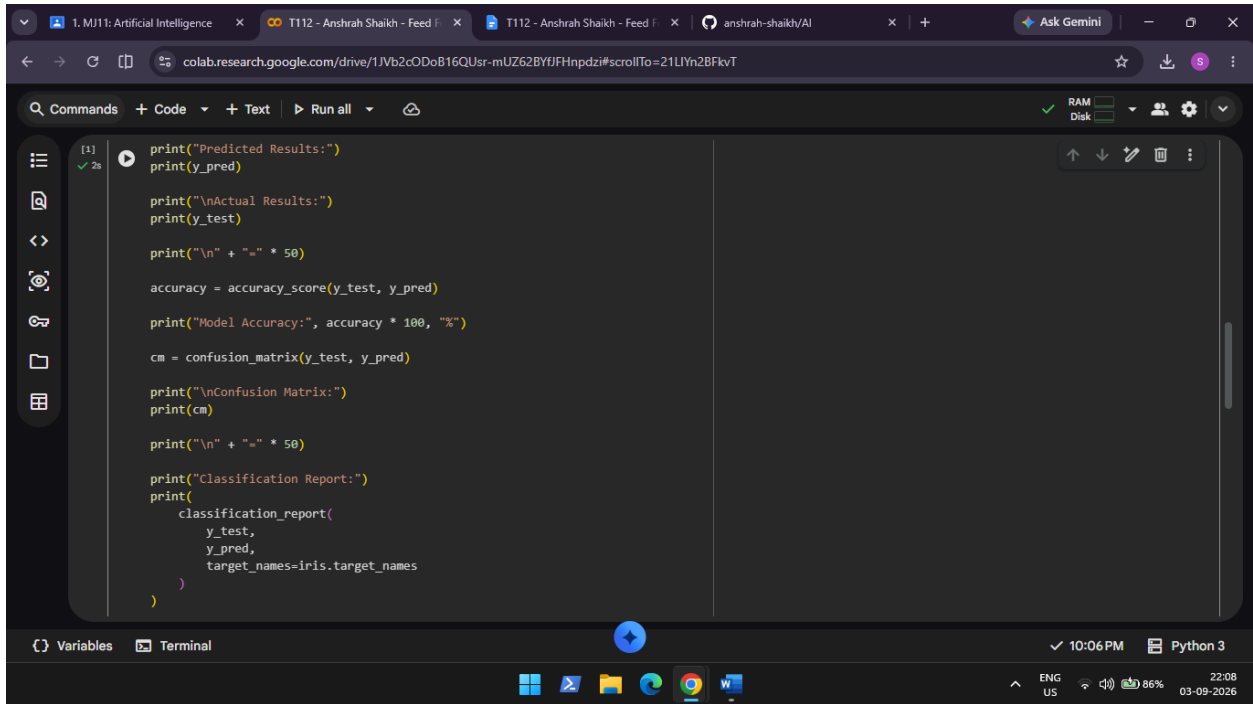
print("\n" + "=" * 50)

y_pred = model.predict(X_test)

print("Predicted Results:")
print(y_pred)
```

ANSHRAH SHAIKH
T112
TYCS
AI PRAC 4

MVLU COLLEGE



The screenshot shows a Google Colab notebook interface. The code cell contains the following Python code:

```
[1]: print("Predicted Results:")
    print(y_pred)

    print("\nActual Results:")
    print(y_test)

    print("\n" + "=" * 50)

    accuracy = accuracy_score(y_test, y_pred)

    print("Model Accuracy:", accuracy * 100, "%")

    cm = confusion_matrix(y_test, y_pred)

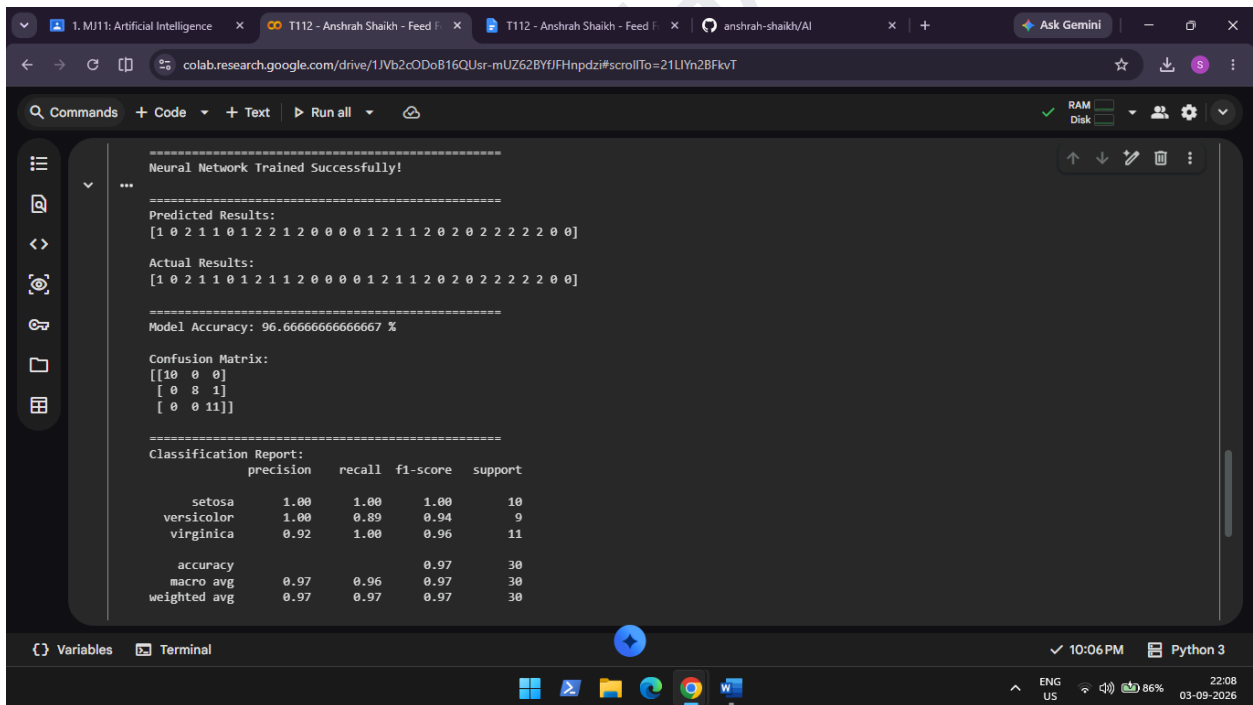
    print("\nConfusion Matrix:")
    print(cm)

    print("\n" + "=" * 50)

    print("Classification Report:")
    print(classification_report(
        y_test,
        y_pred,
        target_names=iris.target_names
    ))
```

The interface includes a top bar with tabs for "1. MJ11: Artificial Intelligence", "T112 - Anshrah Shaikh - Feed F", "T112 - Anshrah Shaikh - Feed F", and "anshrah-shaikh/AI". The bottom status bar shows "10:06 PM", "Python 3", and system icons for language (ENG US), volume, and battery (86%).

Output:



The screenshot shows the output of the code cell from the previous image. The output is as follows:

```
=====
Neural Network Trained Successfully!
=====
Predicted Results:
[1 0 2 1 1 0 1 2 2 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

Actual Results:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

=====
Model Accuracy: 96.6666666666667 %

Confusion Matrix:
[[10  0  0]
 [ 0  8  1]
 [ 0  0 11]]

=====
Classification Report:
      precision    recall  f1-score   support

   setosa         1.00      1.00      1.00        10
  versicolor         1.00      0.89      0.94         9
   virginica         0.92      1.00      0.96        11

   accuracy          0.97      0.96      0.97        30
  macro avg          0.97      0.96      0.97        30
 weighted avg          0.97      0.97      0.97        30
```

The interface is identical to the first screenshot, showing the same tabs and status bar.

ANSHRAH SHAIKH
T112
TYCS
AI PRAC 4